

Term Project — High-Doppler OFDM Receiver

Spring 2026 · 60% of course grade · Individual work

What This Briefing Covers

PART 1 — Project Scope and Impairments (you are here)

- The mission, scope rules, frame parameters
- The four impairments your receiver must beat
- How the BER sweep across Doppler is graded

PART 2 — Implementation, Deliverables and Grading (next)

- Starter pack, TX/RX block diagrams
- What you build, allowed TX changes
- Deliverables, grading rubric, presentation Q&A

You leave today knowing the what, the how, and the rubric.

Project at a Glance

WEIGHT

60% of course grade

FORMAT

Individual — no groups

SUBMISSION

**June 11, 2026
23:59**

PRESENTATION

**June 12, 2026 · 14:30
in person or online**

SEND TO

ecavus@aybu.edu.tr

DURATION

**~10 min talk
+ 5 min Q&A**

Two dates that matter — the email deadline and the talk.

The Mission

WHAT YOU ARE GIVEN

- TX + multipath channel + AWGN as a single black-box MATLAB function.
- One OFDM frame = 10 OFDM symbols.
- QPSK only, uncoded.
- Doppler grid sweep:
 $f_d \in \{0, 1000, 2000, 3000\}$ Hz.
- Residual STO $\leq \pm 32$ samples,
Residual CFO $\leq \pm 12$ kHz.

WHAT YOU BUILD

- Your receiver function:
`student_rx(rx_samples, params)`.
- Estimates and removes residual STO, CFO and Doppler.
- Recovers the QPSK bits.
- Target: lowest BER at lowest MATLAB op-count.
- Beats the Tier-1 baseline at $f_d = 3000$ Hz (the operating point).

TX → **channel** → **STO + CFO + Doppler + AWGN** → **YOU** → **bits**

The TX is fixed. The receiver is yours.

Scope — In vs. Out

In scope	Out of scope
One OFDM frame (10 symbols)	Anything outside this frame
QPSK only, uncoded	Higher-order mod, FEC / channel coding
Data + pilot subcarriers	Preamble, sync training
FPV_C drone multipath channel	Other channel profiles
Custom STO + CFO + Doppler in RX	Modifying the channel
Symbol-level / slot-level interleaving	Channel coding (FEC) of any kind
Pilot pattern rearrangement (≤ 64 pilots/sym)	More than 64 pilot SCs per symbol

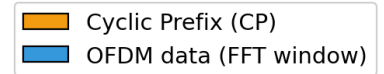
One frame, QPSK only, no FEC — everything else is on the table.

Frame Parameters

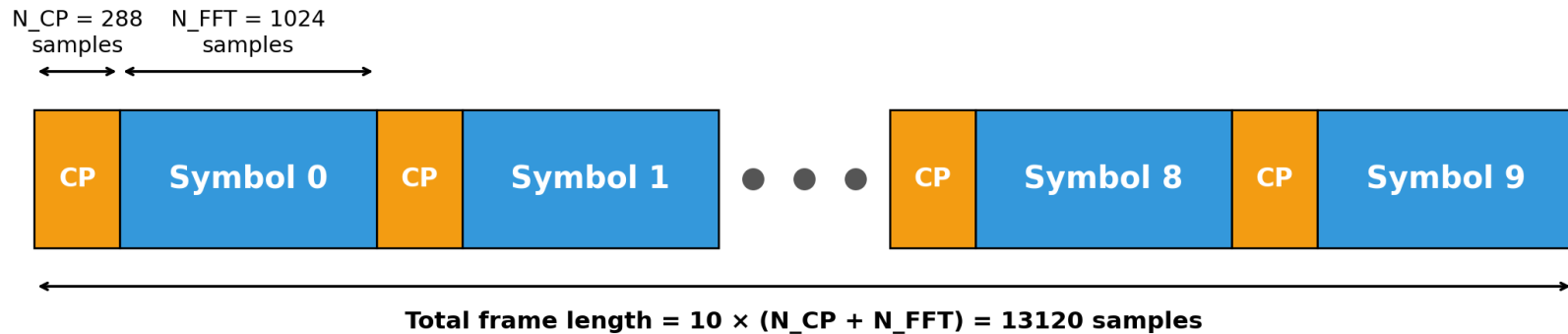
Symbol	Description	Value
Fs	Sampling frequency	30.72 MHz
SCS	Sub-carrier spacing	30 kHz
N_FFT	FFT size	1024 samples
N_CP	CP length	288 samples
N_sym	OFDM symbols per frame	10
BW	Occupied bandwidth	10 MHz
Used SCs	Active SCs per OFDM symbol	312 (26 RB × 12 SC)
Pilots	Pilot SCs per OFDM symbol	64 (dense_4sym pattern)

Sizes are in samples — divide by Fs whenever you need seconds.

One OFDM Frame — Structure



Frame structure — one OFDM frame contains 10 symbols

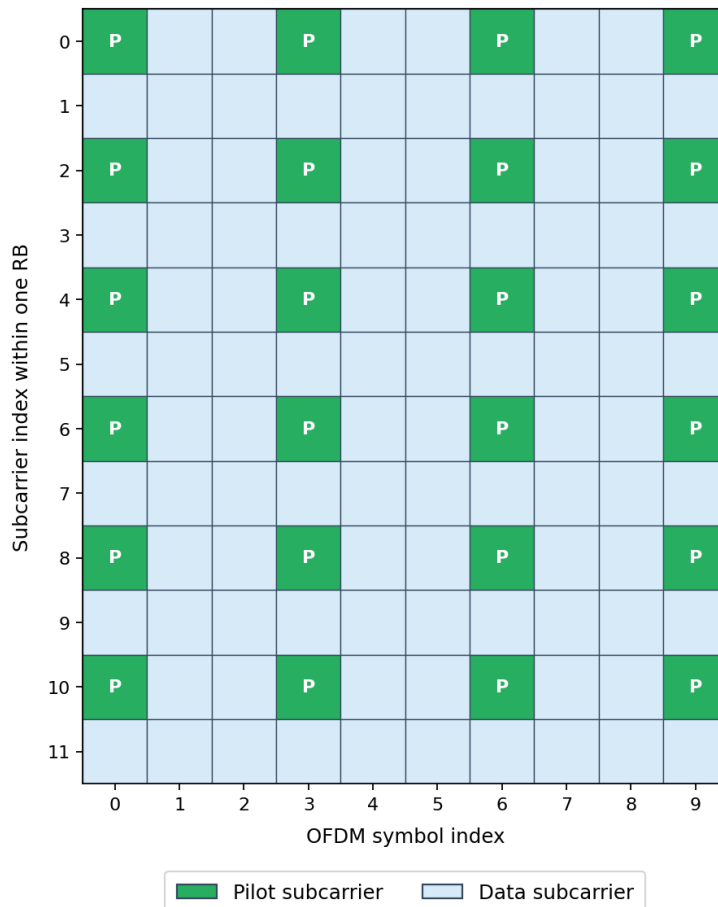


10 OFDM symbols $\times$ (288 CP + 1024 FFT) samples = 13 120 samples per frame.

13 120 complex samples in — your bits out.

Pilot Pattern (dense_4sym)

Pilot pattern (dense_4sym) — one resource block of 12 subcarriers



Each of the 26 resource blocks (312 active subcarriers total) repeats this pattern.
Pilots: 6 SCs × 4 symbols = 24 per RB → 624 pilots per frame.

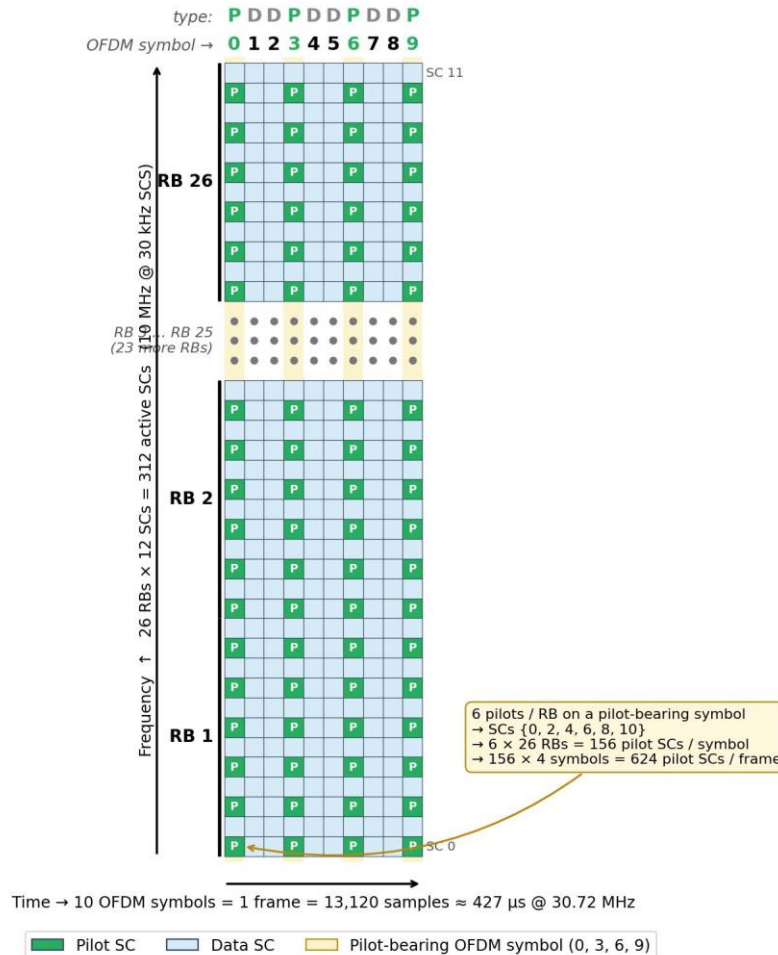
Where pilots sit

- Even SCs of each resource block:
 $\{0, 2, 4, 6, 8, 10\}$
- Symbol indices:
 $\{0, 3, 6, 9\}$
- Total:
64 pilot SCs / OFDM symbol
× 4 pilot symbols per frame
- These are your only known signal — guard them carefully.

Pilots are your only known signal — guard them carefully.

Pilot Pattern — Whole Frame View

Pilot pattern across the full frame — 10 OFDM symbols × 26 resource blocks (dense_4sym)

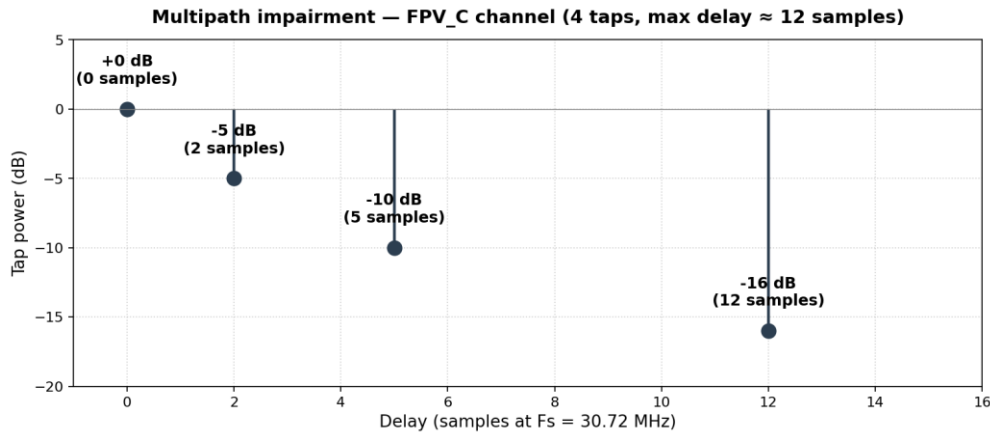


Why this view matters

- Previous slide: 1 RB.
This slide: full 10-sym frame × 26 RBs.
- Yellow columns = 4 pilot symbols (0, 3, 6, 9). 6 P-cells per RB.
- $6 \times 26 \text{ RBs} = 156 \text{ pilot SCs per pilot symbol} \rightarrow 624 \text{ per frame.}$
- Three RBs drawn; the other 23 are identical — the pattern tiles in frequency.

Same dense_4sym tile repeats across every RB — channel-estimator's input is 4 sparse pilot snapshots in time, dense in frequency.

Impairment — Multipath + Doppler



FPV_C channel

- 4 taps, ~ 12 -sample max excess delay.
- `comm.RayleighChannel`, Jakes spectrum.

Doppler grid:

$f_d \in \{0, 1000, 2000, 3000\}$ Hz

- 3000 Hz = operating point (leaderboard score uses this curve).
- 0 Hz collapses to a static channel — sanity check for the RX chain.
- 1 / 2 kHz separate Doppler from STO / CFO in your residual BER.

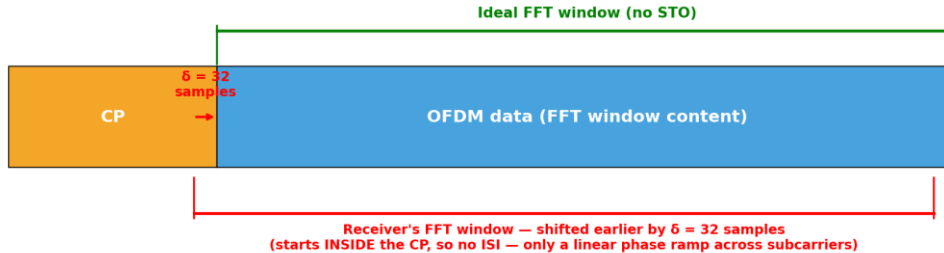
TX override:

`fd_doppler` in `opts` struct.

Operating point is 3 kHz — the lower curves tell you what's hurting you.

Impairment — Residual STO (± 32 samples)

STO impairment — receiver's FFT window starts δ samples earlier, inside the CP



The "ladder" signature

- Instructor sets the value inside `tx_and_channel.m` — you never see it.
- Always within the CP
→ no ISI from the previous symbol.
- Visible effect:
a linear phase ramp across subcarriers (Lec 9 ladder).

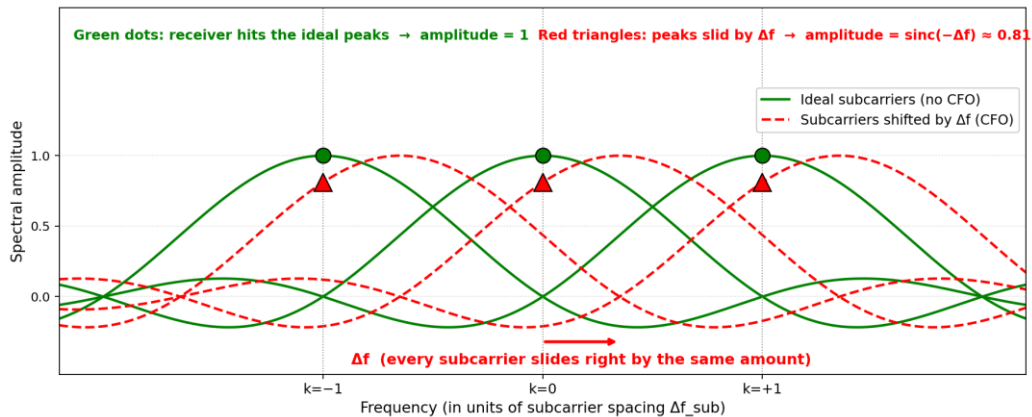
Estimator options:

- unwrap → polyfit (Lab 9), or
- absorb it into the $H[k]$ estimate.

Inside the CP, STO is a phase slope you can fit and remove.

Impairment — Residual CFO (± 12 kHz)

CFO impairment — receiver samples at integer SCs, but the peaks have slid by Δf



Constant over the frame

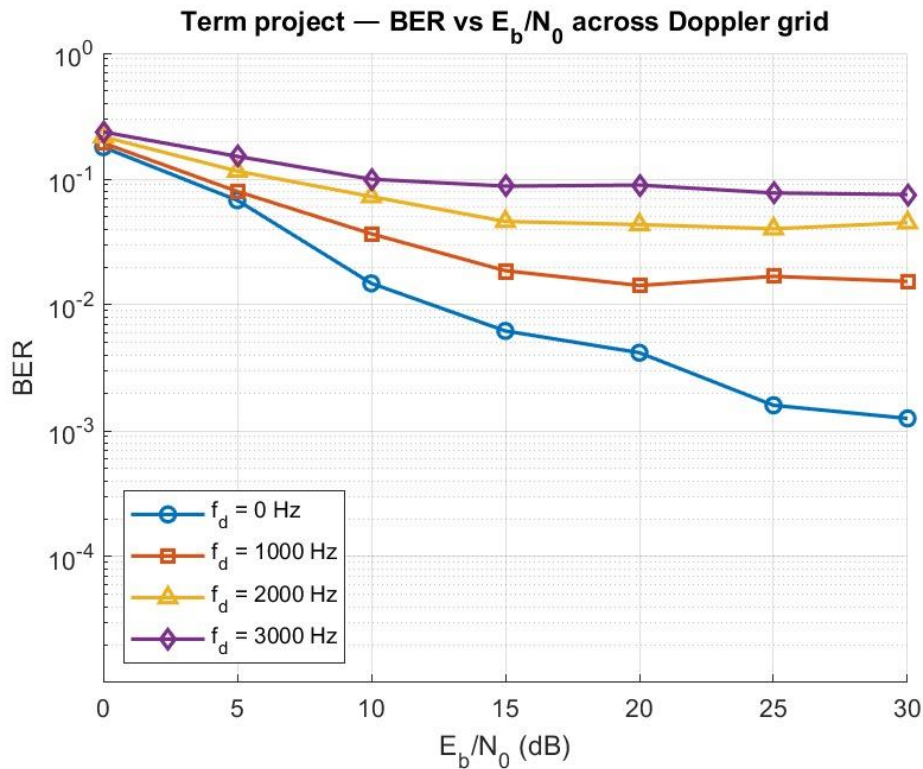
- Every sample rotates by Δf .
Constant across all 10 symbols.
- No preamble in this project $\rightarrow$ full ± 12 kHz HW offset hits you.
- Uncorrected: LS $H[k]$ estimate goes stale in one symbol.

Pick from Lec 10/11:

- M1 — pilot-aided estimator
- M2 — CP-based estimator
- M3 — decision-directed loop
- combine if it helps.

Without CFO correction, your LS channel estimate dies in one symbol.

Baseline BER vs. Doppler



WHAT YOU SEE

$f_d = 0$ keeps falling ($\sim 10^{-3}$). $f_d \geq 1$ kHz hits an irreducible Doppler floor — that floor is the channel-tracker, not STO/CFO.

ADAPTIVE MONTE CARLO

Per E_b/N_0 point: stop at ≥ 200 errors OR 5000 frames.

YOUR DEV LOOP

20–50 frames / point is fine while iterating.

FINAL BER PLOT

All 4 Doppler curves + Tier-1 baseline @ $f_d = 3000$ Hz. ≥ 100 errors / point.

Leaderboard is the 3 kHz curve; the others diagnose where the residuals come from.